Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	3 MARKS

3.4 – Technology Stack

The increasing availability of data analytics tools has enabled educational institutions to better understand student behaviour and make informed decisions that enhance student well-being. This case study, *Technology Stack* ***Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study***, examines how a modern technology stack can be leveraged to analyze the dietary habits and food preferences of college students. The study utilizes a comprehensive data pipeline consisting of data collection platforms, relational databases, data cleaning and transformation tools, and Tableau for advanced visualization and reporting. Data is gathered from student surveys, cafeteria transaction records, nutritional databases, and demographic information, providing a holistic view of students' eating patterns. After preprocessing and integrating the data, Tableau dashboards are developed to visualize key metrics such as meal frequency, food category preferences, calorie consumption, spending behaviour, and the relationship between dietary choices and factors such as academic workload, living arrangements, and income levels.

Through interactive visualizations and analytical dashboards, Tableau enables stakeholders to identify trends and uncover actionable insights. For example, the analysis may reveal that students with busy academic schedules tend to consume more fast food, while those living on campus are more likely to choose balanced meals due to easier access to dining facilities. The dashboards also highlight nutritional gaps, such as insufficient fruit and vegetable intake or excessive consumption of sugary beverages. These findings support the development of targeted dietary strategies, including nutrition awareness campaigns, personalized meal recommendations, healthier cafeteria menu options, and cost-effective meal plans. Additionally, predictive analytics and trend monitoring can help institutions anticipate future dietary needs and evaluate the effectiveness of implemented interventions. By combining a robust technology stack with Tableau's powerful visualization capabilities, the study demonstrates how data-driven approaches can improve dietary decision-making, enhance student health outcomes, and support the creation of a healthier campus environment. Ultimately, this case study illustrates the value of integrating technology and analytics to transform raw food consumption data into meaningful insights that drive positive behavioural and institutional change.